

Active Directory User Lifecycle & Access Management

Area of Focus Active Directory / Identity & Access Management

Project Overview

This project demonstrates the administration of the Active Directory user lifecycle within a Windows Server 2022 domain environment for the fictional organization LOG(N) Pacific.

The project simulates common Systems Administrator and IT support responsibilities associated with managing employee identities and access throughout their employment lifecycle. Tasks include creating and configuring user accounts, assigning group-based departmental access, performing password resets and account unlocks, modifying access following employee role changes, disabling and relocating accounts during offboarding, and auditing Active Directory accounts and security-group memberships.

PowerShell and Active Directory administrative tools are used to verify account status, group membership, organizational unit placement, and access permissions. The project also includes a simulated Tier 2 identity and access management incident, requiring systematic troubleshooting to identify the root cause of an employee access problem, restore the appropriate permissions, and validate the resolution.

The project represents a small-business Active Directory environment and demonstrates how centralized identity and access management can be used to securely manage employee access while supporting common onboarding, support, transfer, offboarding, auditing, and troubleshooting processes.

Key Skills Demonstrated

- Virtualization
- VMware Workstation Pro
- Isolated virtualized Windows Server lab environment
- Server Environment
- Microsoft Windows Server 2022
- LOG(N) Pacific Active Directory domain (logn.local)
- Domain Controller (DC01)
- Identity & Access Management
- Active Directory Domain Services (AD DS)
- Active Directory Users and Computers (ADUC)
- Organizational Units (OUs)

- Active Directory user accounts
 - Active Directory security groups
 - Group-based access control
 - User account enable/disable management
 - Password reset and account unlock administration
 - File & Access Management
 - Windows Server File Services
 - SMB network shares
 - NTFS permissions
 - Security-group-based resource access
 - Departmental Finance share (\\DC01\\Finance)
 - Administration & Automation
 - Windows PowerShell
 - Active Directory PowerShell module
 - Get-ADUser
 - Get-ADGroup
 - Get-ADGroupMember
 - Search-ADAccount
 - PowerShell-based identity and access auditing
 - Troubleshooting & Validation
 - Active Directory account-status verification
 - Security-group membership validation
 - OU placement verification
 - SMB share and NTFS permission validation
 - Tier 2 identity/access troubleshooting
 - Root-cause analysis and post-resolution verification
-

Environments & Technologies Used

Virtualization

- VMware Workstation Pro
- Isolated virtualized Windows Server lab environment

Server Environment

- Microsoft Windows Server 2022
- LOG(N) Pacific Active Directory domain (logn.local)
- Domain Controller (DC01)

Identity & Access Management

- Active Directory Domain Services (AD DS)

- Active Directory Users and Computers (ADUC)
- Organizational Units (OUs)
- Active Directory user accounts
- Active Directory security groups
- Group-based access control
- User account enable/disable management
- Password reset and account unlock administration

File & Access Management

- Windows Server File Services
- SMB network shares
- NTFS permissions
- Security-group-based resource access
- Departmental Finance share (\\DC01\\Finance)

Administration & Automation

- Windows PowerShell
- Active Directory PowerShell module
- Get-ADUser
- Get-ADGroup
- Get-ADGroupMember
- Search-ADAccount
- PowerShell-based identity and access auditing

Troubleshooting & Validation

- Active Directory account-status verification
- Security-group membership validation
- OU placement verification
- SMB share and NTFS permission validation
- Tier 2 identity/access troubleshooting
- Root-cause analysis and post-resolution verification

Tasks Performed

- Validated the existing Windows Server 2022 Active Directory domain environment.
- Created and configured a new employee Active Directory user account.
- Placed the employee account within the appropriate departmental Organizational Unit (OU).
- Configured employee information including department and job title.
- Assigned departmental access through Active Directory security-group membership.
- Verified group-based access to departmental network resources.

- Performed an administrative password reset for a user support request.
 - Investigated and resolved an Active Directory account lockout.
 - Simulated an employee department transfer and updated the user's identity information.
 - Removed obsolete departmental access following the employee's role change.
 - Assigned new security-group membership appropriate to the employee's new position.
 - Moved the employee account to the appropriate departmental OU.
 - Performed employee offboarding by disabling an account and removing unnecessary access.
 - Moved the disabled employee account into the designated Disabled Users OU.
 - Audited enabled/disabled accounts and Active Directory security-group memberships.
 - Used PowerShell to query and verify Active Directory users, groups, account status, and access.
 - Created PowerShell-based Active Directory user and access reporting.
 - Performed a simulated Tier 2 identity/access incident investigation.
 - Identified the root cause of the access issue and implemented the appropriate resolution.
 - Verified Active Directory and resource access following remediation.
 - Performed final validation of the Active Directory identity-management environment.
-

Project Workflow

The project followed the complete lifecycle of an employee identity within the LOG(N) Pacific Active Directory environment:

1. Environment Validation

Verified that Active Directory Domain Services, DNS, departmental OUs, security groups, and existing resources were operational before making changes.

2. Employee Onboarding

Created a new Finance employee account, configured the employee's identity information, and placed the account within the appropriate Finance OU.

3. Access Provisioning

Added the employee to the Finance-SG security group and verified that departmental access was being controlled through group membership rather than permissions assigned directly to the individual user.

4. Account Support & Password Administration

Simulated common help-desk requests by performing password administration and resolving an account lockout while verifying the employee's account status.

5. Employee Role Transfer

Simulated an employee moving from Finance to IT. Updated the employee's department and job information, removed obsolete Finance access, assigned appropriate new group membership, and relocated the account to the correct OU.

6. Access Modification & Verification

Verified that access associated with the employee's previous position had been removed and that the account reflected the requirements of the new role.

7. Employee Offboarding

Simulated the departure of an employee by disabling the account, removing departmental access, documenting the account status, and moving the account into the **Disabled Users OU** rather than immediately deleting it.

8. Identity & Access Audit

Reviewed Active Directory accounts and security-group memberships to identify disabled accounts and verify that departmental access remained appropriately assigned.

9. PowerShell Administration & Reporting

Used PowerShell to query Active Directory and automate portions of user, account-status, group-membership, and access reporting.

10. Tier 2 Break/Fix Troubleshooting

Investigated a simulated employee access problem using a structured troubleshooting process. Checked account status, OU placement, security-group membership, and resource permissions to isolate the root cause before implementing a corrective action.

11. Resolution & Verification

Restored the appropriate access and verified that the affected account and resources operated as expected following remediation.

12. Final Environment Validation

Performed final checks of Active Directory services, user accounts, organizational structure, security groups, and access configuration to confirm the identity-management environment remained operational.

Workflow Summary

Onboard → Provision Access → Support → Transfer → Modify Access → Offboard → Audit → Automate → Troubleshoot → Validate

Screenshots & Explanations

Active Directory Environment Validation

```
PS C:\Users\Administrator> Get-Service ADWS,DNS |
>> Select-Object Name,Status

Name      Status
-----
ADWS      Running
DNS       Running

PS C:\Users\Administrator>
PS C:\Users\Administrator> Get-ADDomain

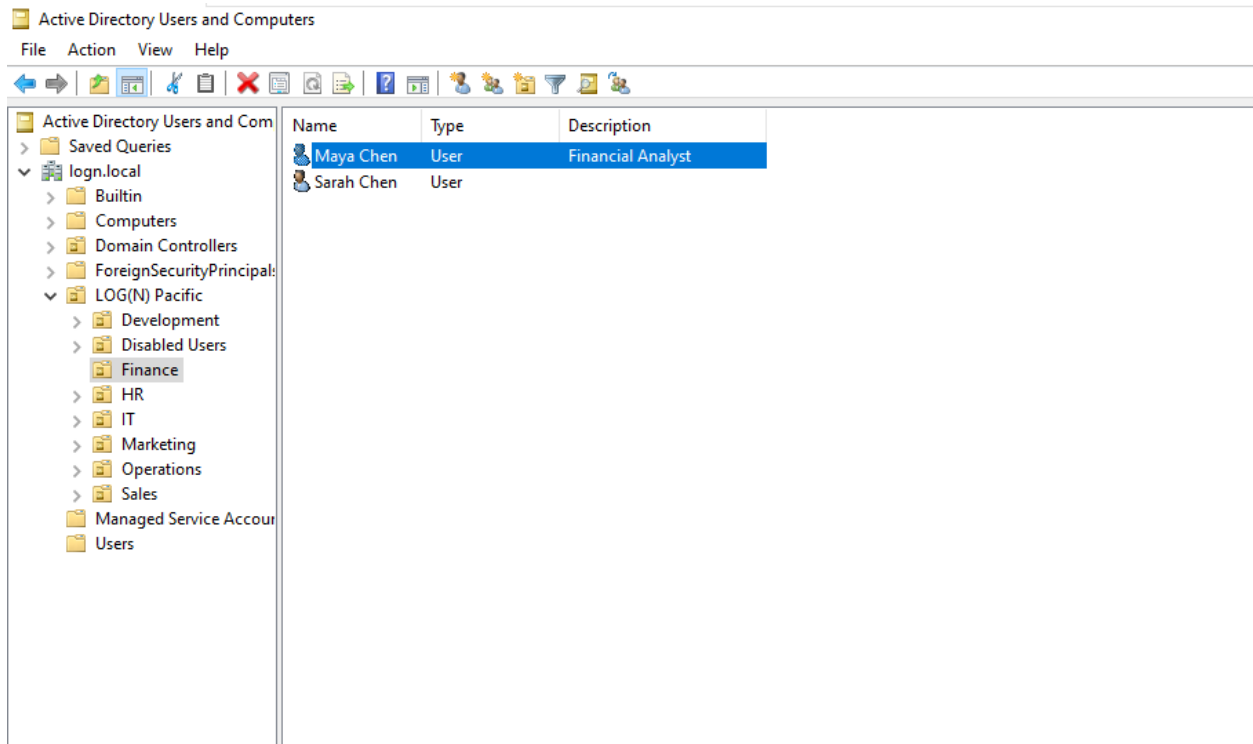
AllowedDNSSuffixes      : {}
ChildDomains            : {}
ComputersContainer      : CN=Computers,DC=logn,DC=local
DeletedObjectsContainer : CN=Deleted Objects,DC=logn,DC=local
DistinguishedName       : DC=logn,DC=local
DNSRoot                 : logn.local
DomainControllersContainer : OU=Domain Controllers,DC=logn,DC=local
DomainMode              : Windows2016Domain
DomainSID               : S-1-5-21-2431518124-1919275917-343495264
ForeignSecurityPrincipalsContainer : CN=ForeignSecurityPrincipals,DC=logn,DC=local
Forest                 : logn.local
InfrastructureMaster    : DC01.logn.local
LastLogonReplicationInterval :
LinkedGroupPolicyObjects : {CN={3182F340-016D-11D2-945F-00C04FB984F9},CN=Policies,CN=System,DC=logn,DC=local}
LostAndFoundContainer   : CN=LostAndFound,DC=logn,DC=local
ManagedBy              :
Name                    : logn
NetBIOSName             : LOGN
ObjectClass              : domainDNS
ObjectGUID               : e49126bb-bbd2-4651-8a36-1333d31672d3
ParentDomain            :
PDCemulator             : DC01.logn.local
PublicKeyRequiredPasswordRolling : True
QuotasContainer         : CN=NTDS Quotas,DC=logn,DC=local
ReadOnlyReplicaDirectoryServers : {}
ReplicaDirectoryServers : {DC01.logn.local}
RIDMaster               : DC01.logn.local
SubordinateReferences   : {DC=ForestDnsZones,DC=logn,DC=local, DC=DomainDnsZones,DC=logn,DC=local,
                          CN=Configuration,DC=logn,DC=local}
SystemsContainer        : CN=System,DC=logn,DC=local
UsersContainer          : CN=Users,DC=logn,DC=local

PS C:\Users\Administrator>
PS C:\Users\Administrator> Get-ADGroup -Identity "Finance-SG"

DistinguishedName : CN=Finance-SG,OU=LOG(N) Pacific,DC=logn,DC=local
GroupCategory     : Security
GroupScope        : Global
Name              : Finance-SG
ObjectClass       : group
ObjectGUID        : cb135904-8959-4bb6-af96-9336dd17cfad
SamAccountName    : Finance-SG
SID               : S-1-5-21-2431518124-1919275917-343495264-1110

PS C:\Users\Administrator>
```

New Finance Employee Account



Group-Based Departmental Access

```
PS C:\Users\Administrator>
PS C:\Users\Administrator> Get-ADGroupMember "Finance-SG" |
>> Select-Object Name,SamAccountName

Name          SamAccountName
----          -
Sarah Chen    schen
Maya Chen     mchen

PS C:\Users\Administrator> _
```

Group-Based Finance Resource Authorization

```
PS C:\Users\Administrator> Get-ADGroupMember "Finance-SG" | Select-Object Name,SamAccountName

Name          SamAccountName
-----
Sarah Chen    schen
Maya Chen     mchen

PS C:\Users\Administrator> Get-SmbShareAccess -Name "Finance"

Name      ScopeName AccountName      AccessControlType AccessRight
-----
Finance *          LOGN\Domain Admins Allow              Full
Finance *          LOGN\Finance-SG  Allow              Change

PS C:\Users\Administrator> icacls "C:\CompanyData\Finance"
C:\CompanyData\Finance LOGN\Finance-SG:(OI)(CI)(M)
                        NT AUTHORITY\SYSTEM:(I)(OI)(CI)(F)
                        BUILTIN\Administrators:(I)(OI)(CI)(F)
                        BUILTIN\Users:(I)(OI)(CI)(RX)
                        BUILTIN\Users:(I)(CI)(AD)
                        BUILTIN\Users:(I)(CI)(WD)
                        CREATOR OWNER:(I)(OI)(CI)(IO)(F)

Successfully processed 1 files; Failed processing 0 files
PS C:\Users\Administrator>
```

Active Directory Password Administration

```
PS C:\Users\Administrator> Get-ADUser mchen -Properties PasswordLastSet |
>> Select-Object Name,Enabled,PasswordLastSet

Name          Enabled PasswordLastSet
-----
Maya Chen     True

PS C:\Users\Administrator>
```


Employee Account Offboarding

```
Administrator: Windows PowerShell

PS C:\Users\Administrator> Get-ADUser mwilliams | Select-Object Name,SamAccountName,Enabled
Name                SamAccountName Enabled
-----
Marcus Williams mwilliams      False

PS C:\Users\Administrator> Get-ADPrincipalGroupMembership mwilliams |
>> Select-Object Name,GroupScope,GroupCategory
Name                GroupScope GroupCategory
-----
Domain Users        Global      Security

PS C:\Users\Administrator> Get-ADUser mwilliams -Properties Department,Description |
>> Select-Object Name,SamAccountName,Enabled,Department,Description,DistinguishedName
Name                : Marcus Williams
SamAccountName      : mwilliams
Enabled             : False
Department          : Sales
Description          : Offboarded - Employee Separation
DistinguishedName   : CN=Marcus Williams,OU=Disabled Users,OU=LOG(N) Pacific,DC=logn,DC=local

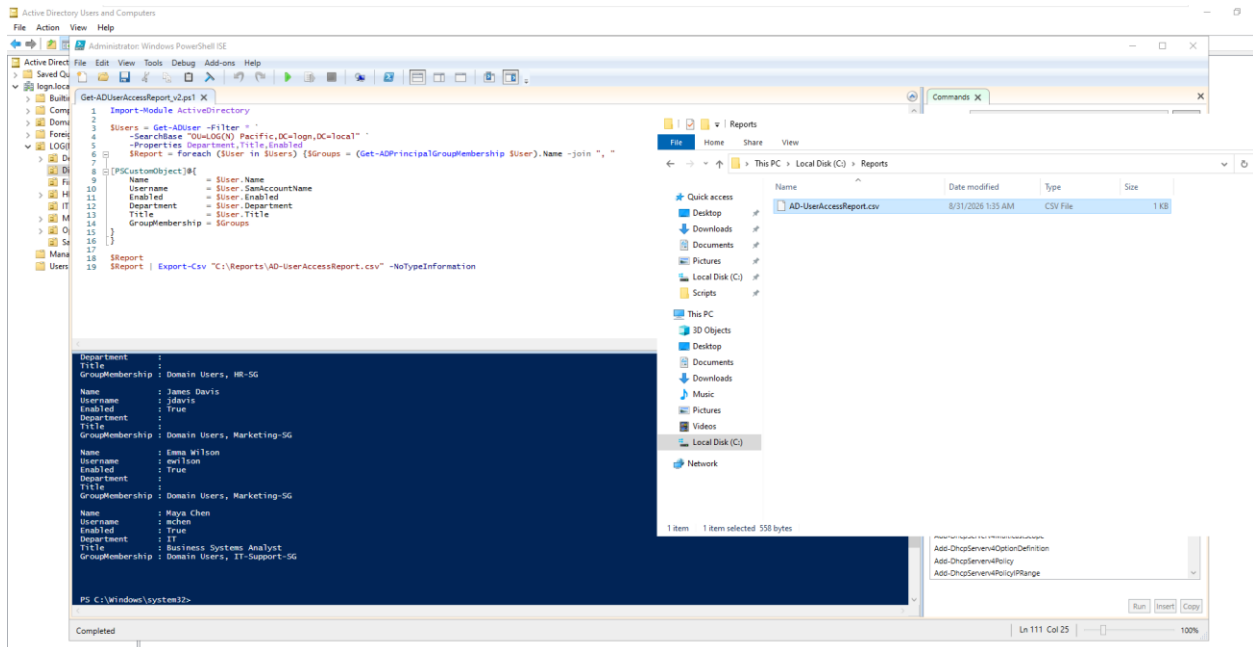
PS C:\Users\Administrator>
```

Active Directory Access & Disabled Account Audit

```
PS C:\Users\Administrator> Get-ADGroupMember "Sales-SG" |
>> Select-Object Name,SamAccountName
>>
>> Get-ADUser -Filter * `
>> -SearchBase "OU=Disabled Users,OU=LOG(N) Pacific,DC=logn,DC=local" `
>> -Properties Department,Enabled |
>> Select-Object Name,SamAccountName,Department,Enabled
Name                SamAccountName Department Enabled
-----
Marcus Williams mwilliams      Sales      False

PS C:\Users\Administrator>
```

PowerShell Active Directory Administration & Reporting



```
1 Import-Module ActiveDirectory
2
3 $Users = Get-ADUser -Filter '*'
4 -SearchBase "OU=LOG(N) Pacific,DC=Togo,DC=local"
5 -Properties Department,Title,Enabled
6 $Report = foreach ($User in $Users) {$Groups = (Get-ADPrincipalGroupMembership $User).Name -join ", "
7
8 [PSCustomObject]@{
9     Name       = $User.Name
10    Username    = $User.DistinguishedName
11    Enabled     = $User.Enabled
12    Department  = $User.Department
13    Title       = $User.Title
14    GroupMemberships = $Groups
15 }
16
17 $Report
18
19 $Report | Export-Csv "C:\Reports\AD-UserAccessReport.csv" -NoTypeInformation
```

Department :
Title :
GroupMemberships : Domain Users, HR-SG
Name : James Davis
Username : jldavis
Enabled : True
Department :
Title :
GroupMemberships : Domain Users, Marketing-SG
Name : Emily Wilson
Username : ewilson
Enabled : True
Department :
Title :
GroupMemberships : Domain Users, Marketing-SG
Name : Maya Chen
Username : mchen
Enabled : True
Department : IT
Title : Business Systems Analyst
GroupMemberships : Domain Users, IT-Support-SG

PS C:\Windows\system32>

[Scenario - INC-002 — IT Department Access Failure] *Employee Unable to Access IT Department Network Resource*

Description

Maya Chen recently transferred from the Finance department to the IT department and was assigned the appropriate IT role and security-group membership. Following the transfer, Maya reported that she could successfully authenticate to the domain but was unable to access the departmental IT network resource required for her new position.

The incident was escalated to Tier 2 for investigation after confirming that her login credentials were functioning correctly.

Tier 2 Troubleshooting — Root Cause Identified

```
PS C:\Users\Administrator> Get-SmbShare | Select-Object Name,Path
>>
>> icacls "C:\CompanyData\IT"

C:\CompanyData\IT NT AUTHORITY\SYSTEM:(I)(OI)(CI)(F)
                  BUILTIN\Administrators:(I)(OI)(CI)(F)
                  BUILTIN\Users:(I)(OI)(CI)(RX)
                  BUILTIN\Users:(I)(CI)(AD)
                  BUILTIN\Users:(I)(CI)(WD)
                  CREATOR OWNER:(I)(OI)(CI)(IO)(F)

Successfully processed 1 files; Failed processing 0 files
Name      Path
----      -
ADMIN$    C:\Windows
C$        C:\
Finance   C:\CompanyData\Finance
IPC$      C:\
NETLOGON  C:\Windows\SYSVOL\sysvol\logn.local\SCRIPTS
SYSVOL    C:\Windows\SYSVOL\sysvol

PS C:\Users\Administrator>
```

Tier 2 Troubleshooting — IT Resource Access Restored

```
Successfully processed 1 files; Failed processing 0 files
Name ScopeName AccountName      AccessControlType AccessRight
-----
IT    *          LOGN\Domain Admins Allow             Full
IT    *          LOGN\IT-Support-SG Allow             Change
True

PS C:\Users\Administrator>
```

A Finance employee reports that they can sign into the domain normally, but they can no longer access the Finance departmental shared folder. Other server functionality appears to be working.

Tier 2 Troubleshooting — Root Cause Identified

```

Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> Get-SmbShare -Name "Finance"

Name      ScopeName Path              Description
-----
Finance *      C:\CompanyData\Finance

PS C:\Users\Administrator> Get-SmbShareAccess -Name "Finance"

Name      ScopeName AccountName      AccessControlType AccessRight
-----
Finance *      LOGN\Domain Admins Allow             Full
Finance *      LOGN\Finance-SG   Allow             Change

PS C:\Users\Administrator> icacls "C:\CompanyData\Finance"
C:\CompanyData\Finance LOGN\Finance-SG:(OI)(CI)(DENY)(RX)
                        NT AUTHORITY\SYSTEM:(OI)(CI)(F)
                        BUILTIN\Administrators:(OI)(CI)(F)
                        BUILTIN\Users:(OI)(CI)(RX)
                        BUILTIN\Users:(CI)(AD)
                        BUILTIN\Users:(CI)(WD)
                        CREATOR OWNER:(OI)(CI)(IO)(F)

Successfully processed 1 files; Failed processing 0 files
PS C:\Users\Administrator>

```

Tier 2 Troubleshooting — Service Restored

```

PS C:\Users\Administrator> icacls "C:\CompanyData\Finance" /grant "LOGN\Finance-SG:(OI)(CI)M"
processed file: C:\CompanyData\Finance
Successfully processed 1 files; Failed processing 0 files
PS C:\Users\Administrator> icacls "C:\CompanyData\Finance"
C:\CompanyData\Finance LOGN\Finance-SG:(OI)(CI)(M)
                        NT AUTHORITY\SYSTEM:(OI)(CI)(F)
                        BUILTIN\Administrators:(OI)(CI)(F)
                        BUILTIN\Users:(OI)(CI)(RX)
                        BUILTIN\Users:(CI)(AD)
                        BUILTIN\Users:(CI)(WD)
                        CREATOR OWNER:(OI)(CI)(IO)(F)

Successfully processed 1 files; Failed processing 0 files
PS C:\Users\Administrator> Get-SmbShareAccess -Name "Finance"
>> Test-Path "\\DC01\Finance"

Name      ScopeName AccountName      AccessControlType AccessRight
-----
Finance *      LOGN\Domain Admins Allow             Full
Finance *      LOGN\Finance-SG   Allow             Change
True

PS C:\Users\Administrator> _

```

Shows the permissions have been restored for the finance department.

Outcome

Successfully administered and validated the complete **Active Directory user lifecycle** within a Windows Server 2022 domain environment for LOG(N) Pacific.

The project demonstrated the management of employee identities from **onboarding through role changes and eventual offboarding**, including user account creation, Organizational Unit

placement, departmental security-group assignments, password administration, account unlocks, access modifications, and disabled-account management.

*Departmental access was managed using **Active Directory security groups, SMB share permissions, and NTFS permissions**, reinforcing a group-based approach to resource authorization rather than assigning permissions directly to individual users.*

PowerShell was used to query Active Directory, verify account and group configurations, perform access audits, and generate an employee identity and access report for administrative review.

*A simulated **Tier 2 access incident** was also investigated using a structured troubleshooting methodology. The investigation confirmed that the affected employee's identity and group membership were correctly configured before identifying missing SMB and NTFS resource permissions as the root cause. The resource was repaired and successfully validated following remediation.*

*The completed environment demonstrated a functional and auditable **identity and access management workflow covering onboarding, access provisioning, employee transfers, offboarding, auditing, automation, troubleshooting, and post-resolution validation.***

What This Project Demonstrates to Employers

This project demonstrates practical experience performing common Systems Administrator and Identity & Access Management responsibilities within a Windows Server Active Directory environment.

It demonstrates the ability to:

- Administer Active Directory user accounts throughout the employee lifecycle, including onboarding, transfers, support, and offboarding.
- Organize and manage identities using Active Directory Organizational Units (OUs).
- Implement group-based access control using Active Directory security groups.
- Provision and revoke departmental resource access as employee responsibilities change.
- Perform common account-support operations including password resets and account unlocks.
- Apply the principle of least privilege by removing access that is no longer required following employee transfers or separation.

- Securely offboard employees by disabling accounts, removing departmental permissions, preserving accounts for administrative purposes, and relocating them to a designated Disabled Users OU.
- Understand the relationship between Active Directory identities, security groups, SMB share permissions, and NTFS permissions.
- Use PowerShell to query users and groups, validate account configuration, perform access reviews, and generate administrative reports.
- Perform Active Directory access audits to identify disabled accounts and validate departmental group membership.
- Troubleshoot Tier 2 identity and resource-access incidents using a structured elimination process rather than making unverified configuration changes.
- Conduct root-cause analysis, implement corrective actions, and perform post-resolution validation.
- Document identity-management activities and technical troubleshooting in a format suitable for operational support and change tracking.